# C PROGRAMMING LANGUAGE OPERATORS

PART 2

# TYPES OF OPERATORS

- Relational operators
- Logical operators
- Bit wise operators
- Conditional operators (ternary operators)
- Increment/decrement operators
- Special operators

# Relational Operators

used to find the relation between two variables. i.e. to compare the values of two variables in a C program.

also known as comparison operators

answerable by True or False or Yes or No

| Symbols | Name |
|---|---|
| < | less than |
| > | greater than |
| <= | less than or equal to |
| >= | greater than or equal to |
| != | Not equal to |
| == | Equal to |

# Analyze the given table using relational operators

| **Example: A = 5 B = 3** | | |
|---|---|---|
| **Operator** | **Expression** | **Answer** |
| < | A < B | |
| > | A > B | |
| <= | A <= B | |
| >= | A >= B | |
| != | A != B | |
| == | A == B | |

# Logical Operators

these operators are used to perform logical operations on the given expressions.

answerable by True or False or Yes or No

| Symbols | Name |
|---|---|
| && | AND |
| \|\| | OR |
| ! | NOT |

# Analyze the given table using logical operators

| TRUTH TABLE | | | |
|---|---|---|---|
| **Expression** | **AND output (&&)** | **OR output (\|\|)** | **NOT (!)** |
| True && True | True | True | !True = False |
| True && False | False | True | |
| False && True | False | True | !False = True |
| False && False | False | False | |

| **Example: A = 5 B = 2** | |
|---|---|
| **Expression** | **Answer** |
| (A < B) && ( B!=A) | |
| (A == B) \|\| (A >= B) | |

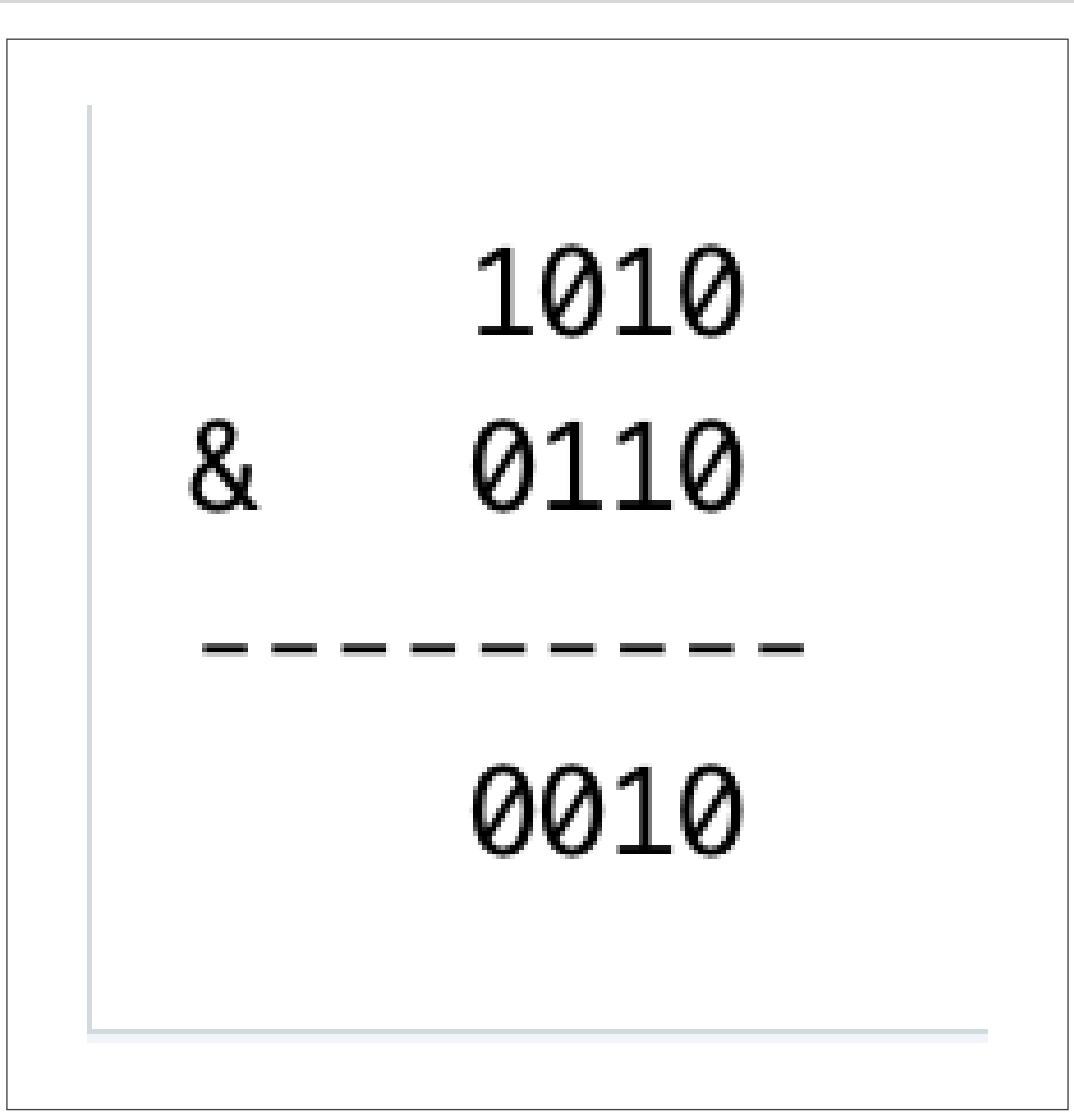
# Analyze the given expression using different types of operators

| **Example: A = 5 B = 2 C = 10** | |
|---|---|
| **Expression** | **Answer** |
| (A*B) < C) \|\| ( C%A == B) | |
| !(((A+C) – B) >= ((C*B)/A)) && (C/B * A <= B*A) | |

# Bitwise Operators

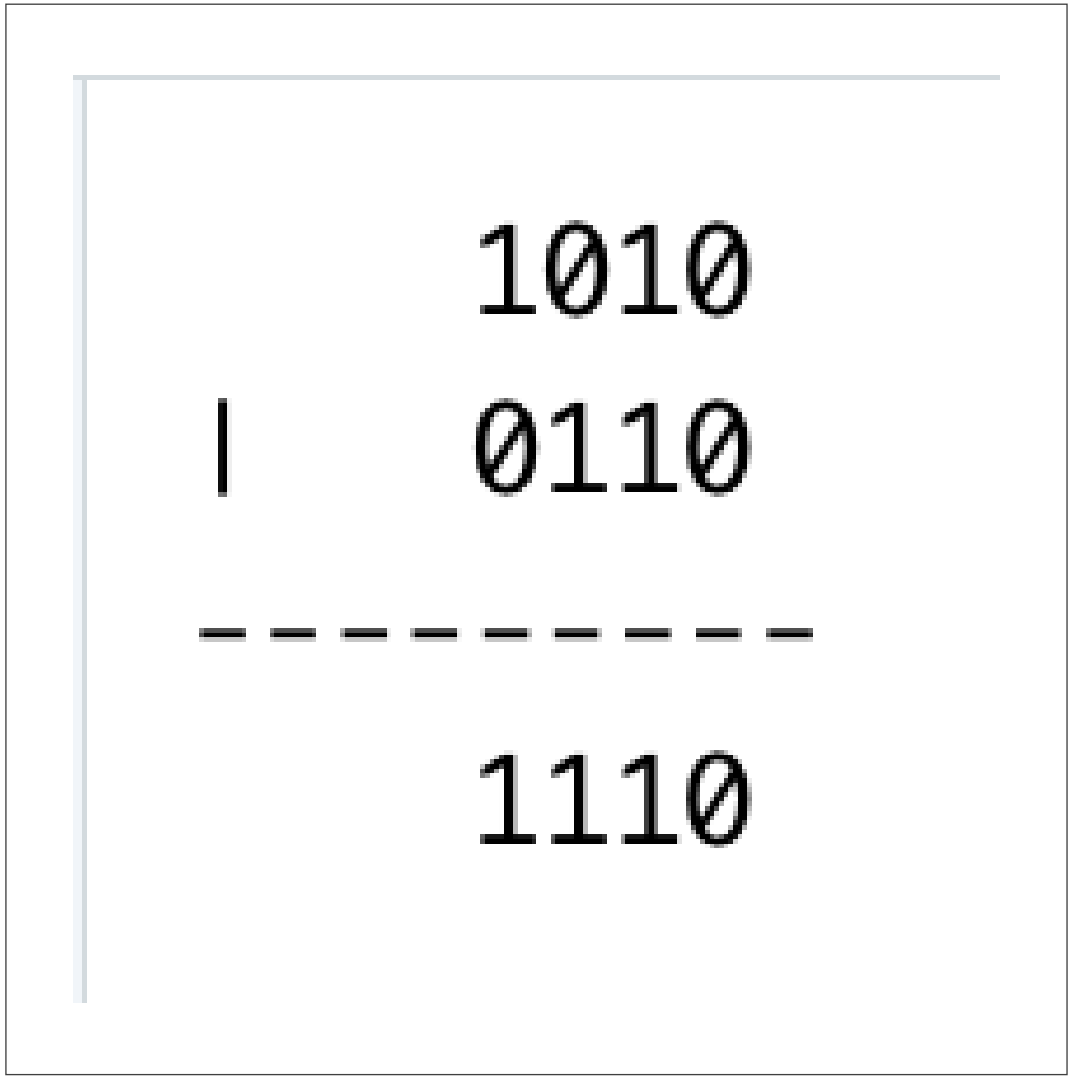
Mathematical operations like addition is converted into bit-level which makes processing faster and saves power.

| Symbols | Name |
|---|---|
| & | Bitwise AND |
| \| | Bitwise OR |
| ^ | Bitwise exclusive OR |
| ~ | Bitwise complement |
| << | Shift left |
| >> | Shift right |



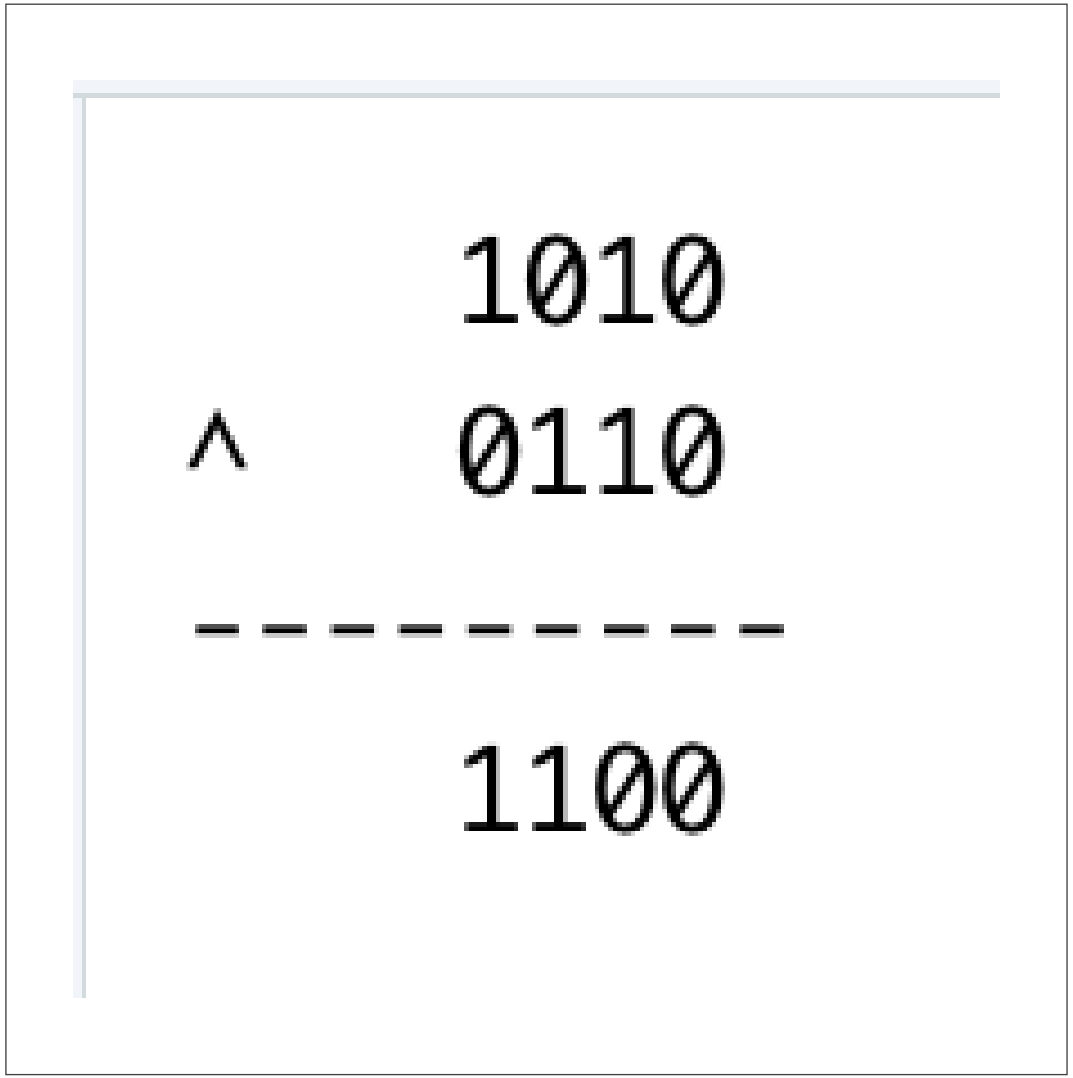

# Bitwise Operators

**AND Operator (&)**: The AND operator compares two bits and returns 1 if both bits are 1, otherwise returns 0.



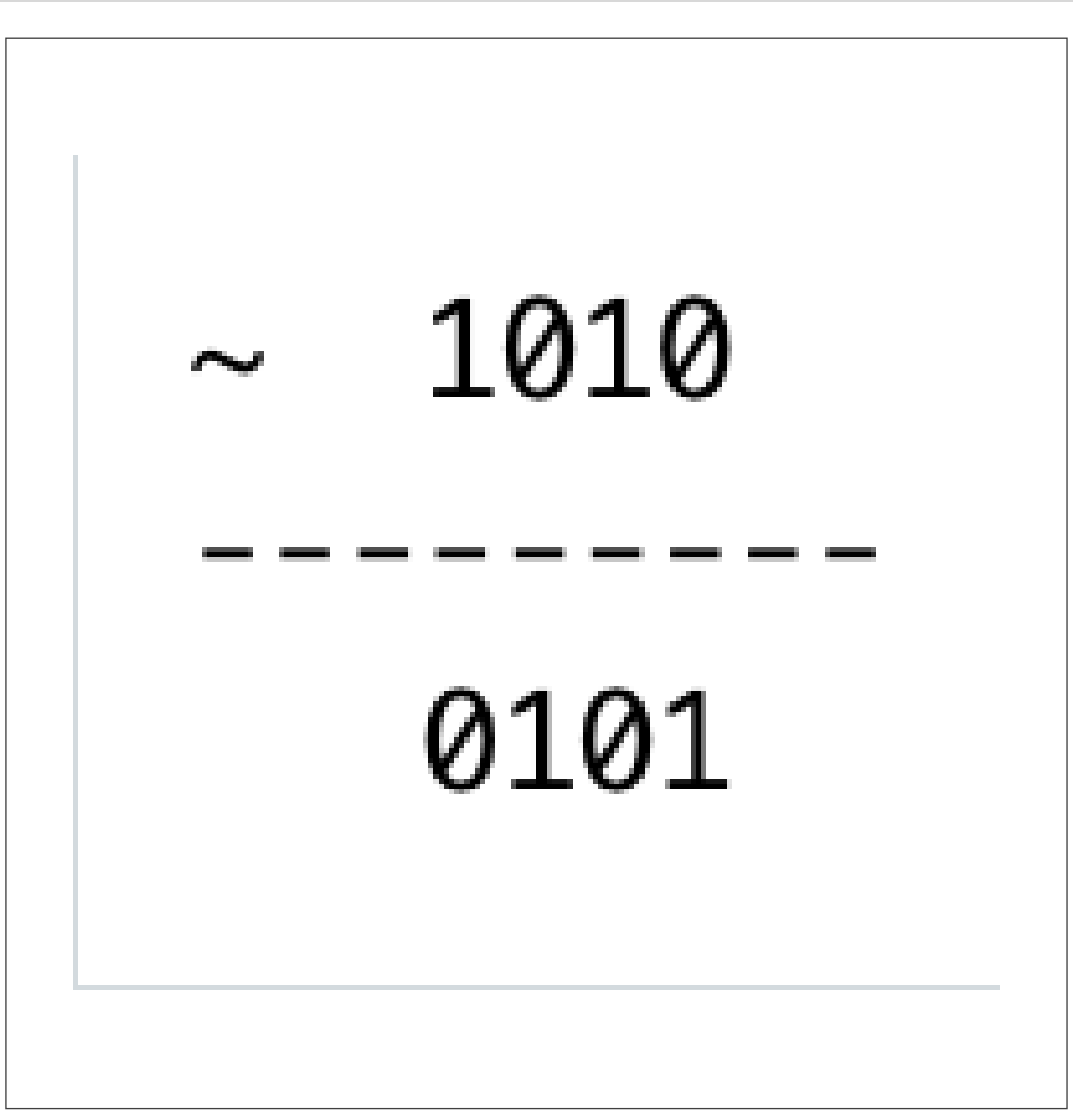

# Bitwise Operators

**OR Operator (|):** The OR operator compares two bits and returns 1 if at least one of the bits is 1, otherwise returns 0



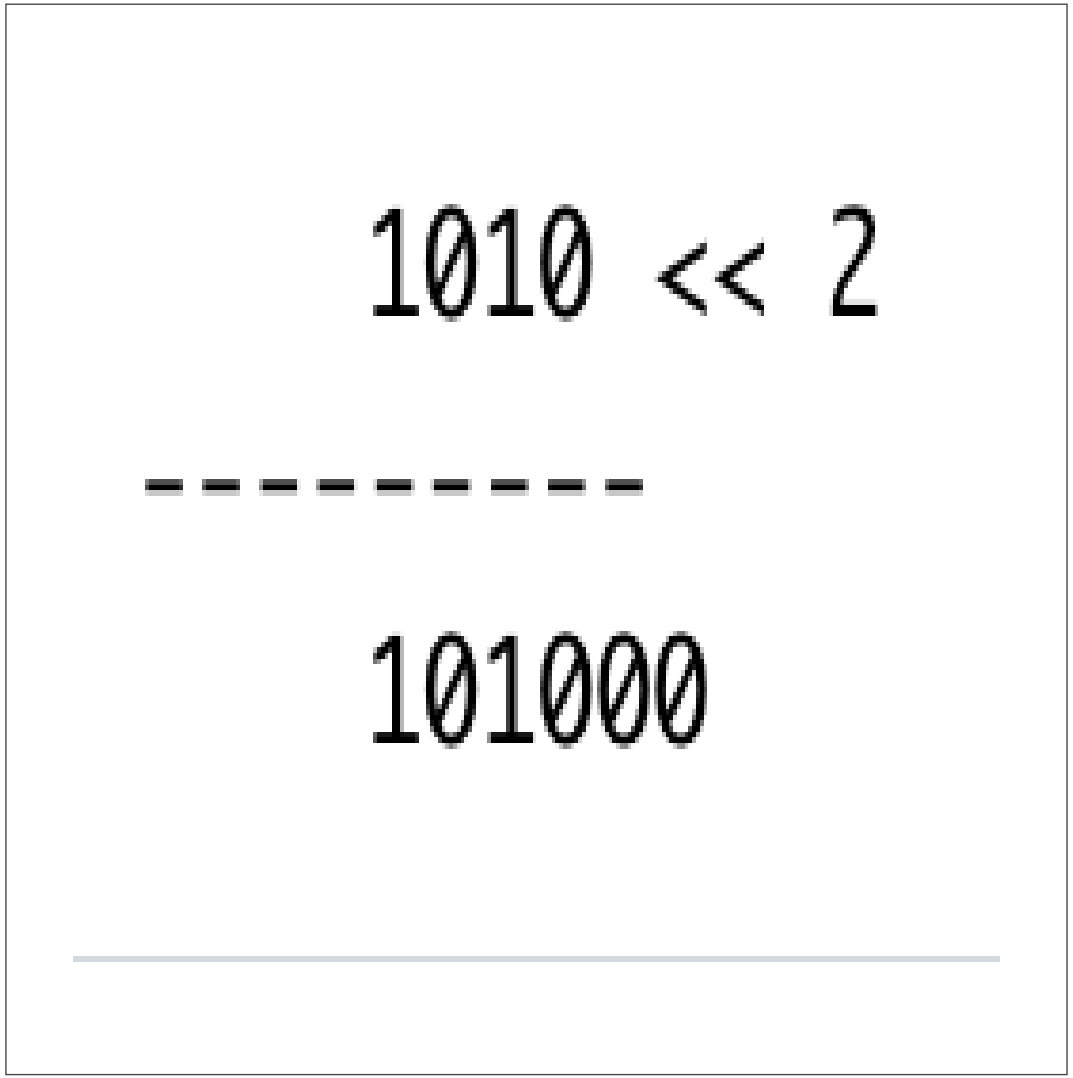

# Bitwise Operators

**XOR Operator (^):** The XOR operator compares two bits and returns 1 if the bits are different (one 0 and the other 1), otherwise returns 0.



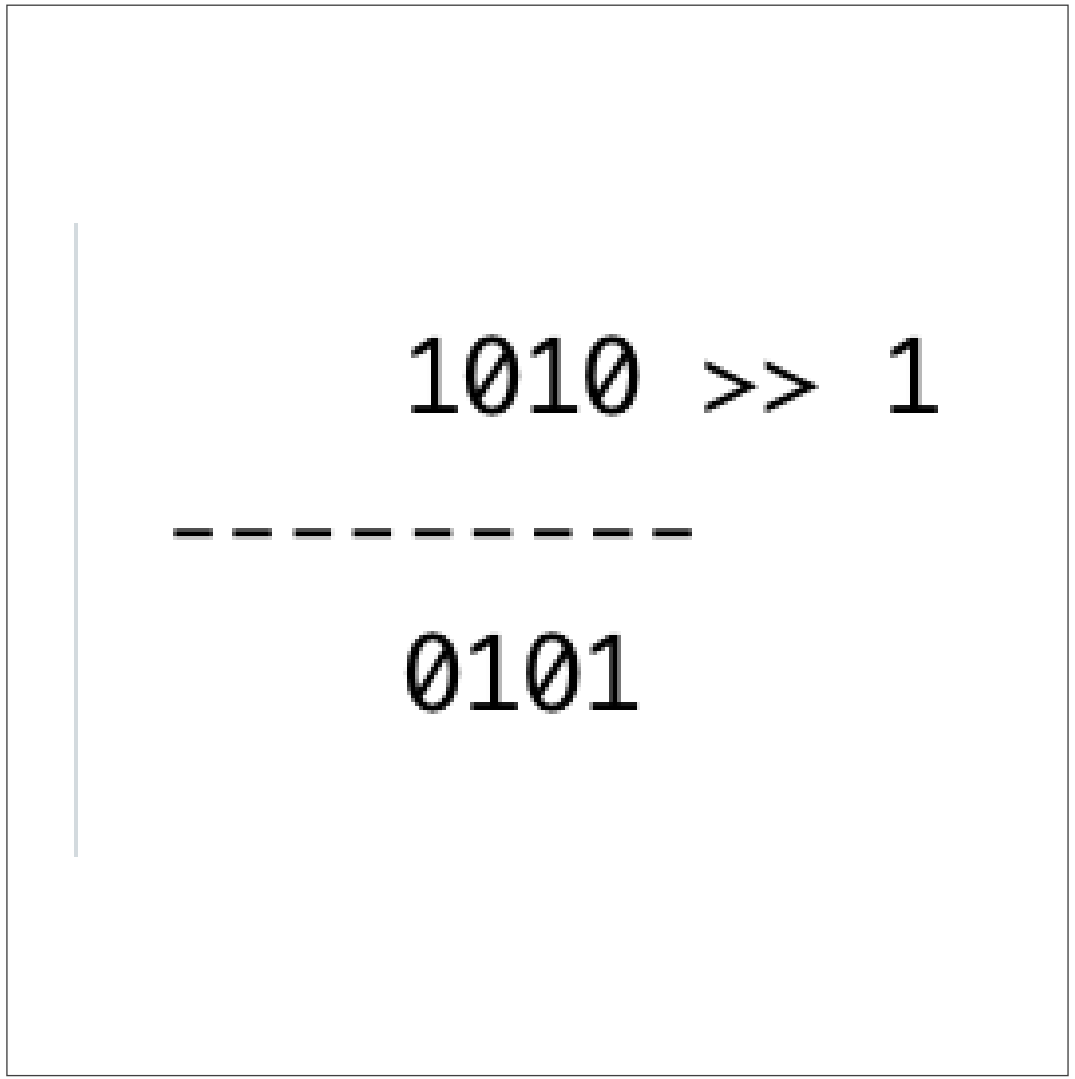

# Bitwise Operators

**Complement Operator (~):** The complement operator flips the bits, turning 0 to 1 and 1 to 0.




# Bitwise Operators

**Left Shift Operator (<<):** The left shift operator shifts the bits of a number to the left by a specified number of positions, effectively multiplying the number by a power of 2




# Bitwise Operators

**Right Shift Operator (>>):** The right shift operator shifts the bits of a number to the right by a specified number of positions, effectively dividing the number by a power of 2

# Increment / Decrement Operators

Mathematical operations like addition is converted into bit-level which makes processing faster and saves power.

| Symbols | Name |
|---|---|
| ++ | Increment |
| -- | Decrement |

# Conditional Operator

This is also known as ternary operator that can work on 3 operands.

- Syntax: conditionalExpression ? expression1 : expression2
- The first expression conditionalExpression is evaluated first.
- This expression  evaluates to 1 if it's true and evaluates to 0 if it's false.
    - If conditionalExpression is true, expression1 is evaluated.
    - If conditionalExpression is false, expression2 is evaluated.

| Symbols | Name |
|---|---|
| ? | Increment |

# Sample Program for Conditional Operator

```
1.  #include <stdio.h>
2.  int main()
3.  {
4.  char February;
5.  int days;
6.  printf("If this year is leap year, enter 1. If not enter any integer: ");
7.  scanf("%c",&February);

8.  // If test condition (February == 'l') is true, days equal to 29.
9.  // If test condition (February =='l') is false, days equal to 28.
10. days = (February == '1') ? 29 : 28;

11. printf("Number of days in February = %d",days);
12. }
```

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